

Project Report: AI-Powered Learning Assistant

Official Title: Emotion Detection & Learning Support Engine

Technical Version: 1.0.0

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1. Project Team

Team Leader

- Vennelaganti Datta Vikram

Team Members

- Elchuri Premsai
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2. Executive Summary

The AI-Powered Learning Assistant is a sophisticated educational platform designed to identify learner emotional states and provide real-time, field-aware mentorship. By merging Transformers (BERT), Recurrent Neural Networks (BiLSTM), and Generative AI (Google Gemini), the system creates a bridge between technical problem-solving and emotional support, ensuring students remain motivated during academic challenges.

3. Languages & Technical Frameworks

The project utilizes a multi-disciplinary stack to handle logic, data, and user experience.

3.1 Programming & Data Languages

- Python: The core engine for logic, deep learning, and API integration.
- SQL: Used for architectural schema design and relational data modeling.
- Markdown: Powers the rich-text UI elements and documentation.
- JSON: The format for API data exchange and model configurations.

- PowerShell: Used for environment management and setup automation.
- CSV: Serves as the primary persistence layer for analytics and training data.

4. System Architecture

The application is built on a Modular Neural Architecture:

1. Emotion Engine (NLP Layer): Processes raw input via a dual-model pipeline. It utilizes a custom 50x Hyper-Boost Logic to ensure 90%+ confidence for explicit emotional triggers.
2. Guidance Engine (GenAI Layer): Takes the detected emotion and academic field to orchestrate a personalized response via Gemini 1.5 Flash.
3. Analytics Layer: Consolidates session history into a 3-column interactive dashboard for student self-reflection.

5. Data Design (ER Model)

The system is designed with full database normalization in mind:

- Users Entity: Tracks email (PK), name, role, and login_count.
- Emotion_Records Entity: Stores record_id (PK), input_text, predicted_emotion, secondary_emotion (using a 15% threshold), confidence_score, ai_response, and timestamp.

6. Technical Innovations

6.1 The 50x Hyper-Boost Algorithm

To provide decisive feedback, the system implements a weight-multiplication layer:

$$P_{\text{boosted}} = \frac{P_{\text{raw}} \times 50.0}{\sum(P_{\text{raw}} \times \text{Multipliers})}$$
 This ensures that when a student uses words like "stuck" or "solved," the confidence score reaches a peak of 93%–98%.

6.2 Mixed-Emotion Detection

Unlike standard classifiers, this system identifies "Secondary Emotions." If a

secondary emotion score exceeds 15%, it is displayed alongside the primary state (e.g., Curious + Confused).

7. Dashboard Features

- Sidebar Controls: Allows the user to switch between 11 academic subjects and toggle AI guidance on/off.
- Current Probability Chart: A real-time bar chart showing the weights of all 5 emotions.
- Confidence Journey: A chronological line chart mapping the student's emotional fluctuations.
- Session Mix: A donut chart showing the overall emotional balance of the learning session.

8. Installation & Run Guide

1. Environment: Create a venv and run `pip install -r requirements.txt`.
2. Initialization: Run `python create_model.py` and `python setup_bert.py`.
3. Execution: Launch using `python -m streamlit run app.py`.

9. Conclusion & Future Scope

The AI Learning Assistant successfully demonstrates that AI is most effective when it is emotionally intelligent. Under the leadership of Vennelaganti Datta Vikram and the development team, this project provides a scalable foundation for modern EdTech.

Future Roadmap:

- Multilingual Support: Expanding detection to 50+ languages.
- Multimodal Analysis: Integrating facial expression and voice tone analysis.
- LMS Integration: Direct plugins for Canvas, Moodle, and Blackboard.